

Global Valuation

Cross-Country Price-Earnings Multiples

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Data and Sample

Firm-Level Regressions

Country Effects Are Large and Persistent

What Predicts Country P/E Effects?

Regional Patterns

Summary and Next Steps

Data and Sample

Data Sources

File	Size	Rows	Cols	Description
compustat_global	2.3 GB	981K	468	Global annual fundamentals (122 countries)
compustat_america	0.8 GB	254K	984	North America fundamentals
returns_global	1.9 GB	10.4M	25	Global monthly security prices

- **Combined panel:** 743,087 firm-years, 72,720 firms, 81 countries (1986–2024)
- All financial data in *local currency* (millions)
- **Market cap (Global):** $cshoi \times prccm$ using $prirow$ (primary security)

Variable Definitions

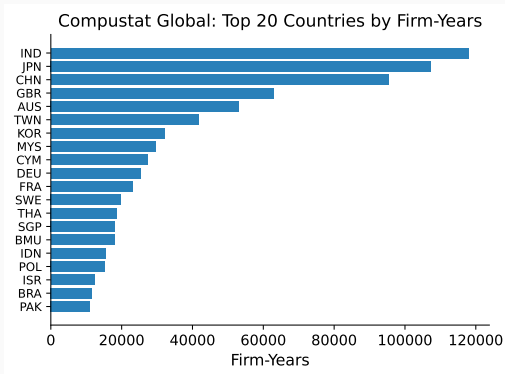
Dependent variable:

$$\log(P/E)_{i,t} = \log\left(\frac{\text{Market Cap}_{i,t}}{\text{Net Income}_{i,t}}\right), \quad NI > 0$$

Market Cap	US: mkvalt. Global: cshoi $\times$ prccm (primary security)
Net Income	ib (income before extraordinary items)
ROA	ib / at
Leverage	(at - ceq)/at
Industry	GICS Industry Group (~25 categories)

P/E restricted to profitable firms (72% of sample). All ratios winsorized at 1st/99th percentiles.

Coverage and Descriptive Statistics



Variable	N	Median	P10	P90
log(P/E)	532K	2.79	1.76	4.32
ROA	743K	0.02	-0.15	0.11
Leverage	743K	0.49	0.14	0.85

Top 5 by obs: USA (188K), JPN (60K), CHN (58K), IND (57K), TWN (31K).

Global prices start 2007.

Firm-Level Regressions

Panel Regression: P/E and Firm Controls

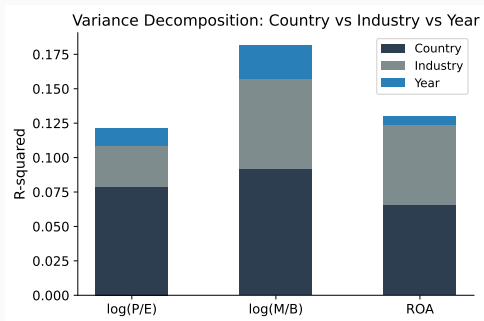
Dep. variable: $\log(P/E)_{i,t}$ (profitable firms, $NI > 0$)

	(1)	(2)	(3)
Leverage	−0.017 (0.102)	−0.013 (0.100)	−0.015 (0.100)
$\log(\text{Assets})$	−0.076** (0.030)	−0.078** (0.031)	−0.075** (0.030)
Lag Δ Earnings	−0.016*** (0.003)	−0.016*** (0.003)	−0.016*** (0.003)
R&D / Assets			+2.98*** (0.33)
Fixed effects	C + I + Y	C×Y + I	C + I + Y
N / R^2	428K / 0.142	428K / 0.163	428K / 0.148

SE clustered by country (in parentheses). C = country, I = industry (GICS), Y = year.

Firm-level R&D is priced ($t \approx 9$) but **country effects barely change**: cross-country std dev of $\hat{\gamma}_c$ drops by only 0.3%.

Variance Decomposition: Country vs Industry vs Year



Shapley R^2 decomposition of $\log(P/E)$:

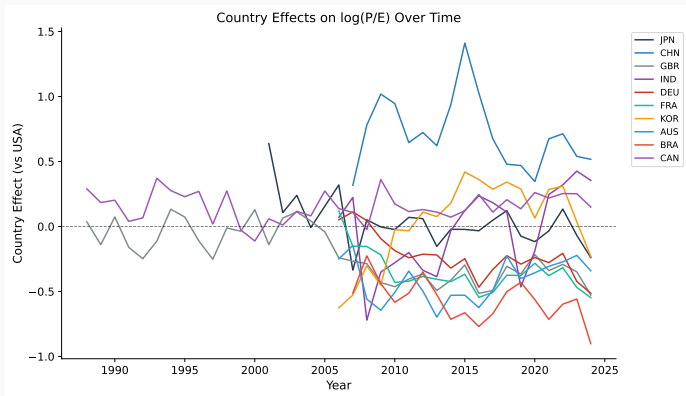
Factor	Share of R^2
Country	64.6%
Industry	24.8%
Year	10.6%

Country alone explains nearly two-thirds of the variation in P/E .

Extends Heston–Rouwenhorst (1994) from *returns* to *valuation levels*.

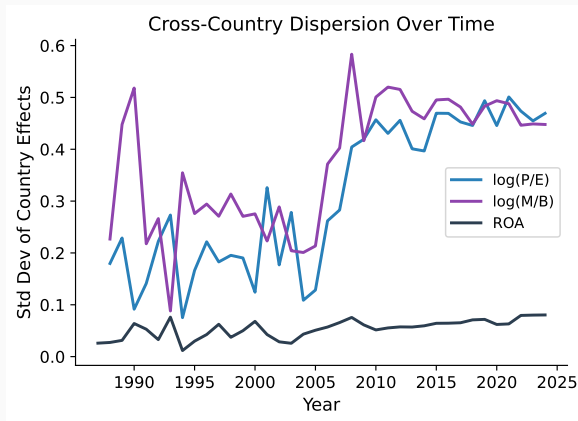
Country Effects Are Large and Persistent

Country P/E Effects Over Time (vs USA)



Year-by-year regressions with size, leverage, earnings growth, industry FE. USA = 0. China premium largest but declining. Japan near zero. India rising since 2015.

P/E Dispersion Is Diverging



P/E dispersion:

$$\sigma = 0.17 \rightarrow 0.47 \quad (\times 2.7)$$

Despite financial globalization, country valuations are *not converging*.

The gap between the most and least expensive countries has nearly tripled since the early 2000s.

What Predicts Country P/E Effects?

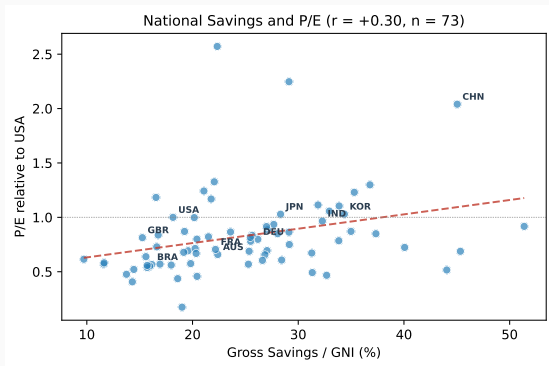
Cross-Country Regressions: P/E Effects on Country Characteristics

Dep. variable: average country P/E effect $\hat{\gamma}_c$ (log scale, USA = 0)

	(1)	(2)	(3)	(4)	(5)	(6)
R&D / Assets	4.86*** (4.73)			4.67*** (4.72)		6.29*** (5.89)
Savings / GNI		0.016*** (3.10)		0.013*** (2.75)		0.014*** (2.88)
IMF GDP growth (2025–29)			0.051 (1.42)		0.051 (1.69)	0.051* (1.69)
R^2	0.23	0.12	0.03	0.35	0.15	0.45
N	76	73	73	65	65	65

t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. R&D and savings both independently significant. Growth useful as control but not significant alone.

National Savings Rate Predicts P/E



Gross savings / GNI (World Bank):

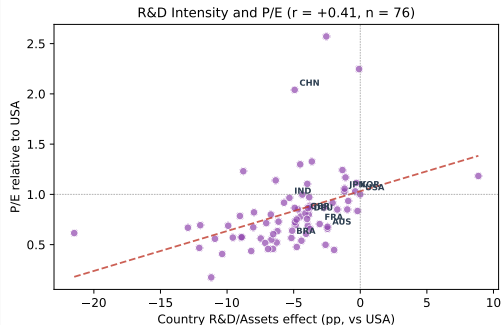
$r = +0.30$, $t = 2.65^{***}$ ($n = 73$)

Robust:

- Excl. Gulf states: $t = 3.40$
- Controlling for R&D: $t = 2.75$
- Controlling for GDP growth: $t = 2.88$

Demand-side channel: more capital seeking equity returns pushes up price multiples.

R&D Intensity Predicts P/E

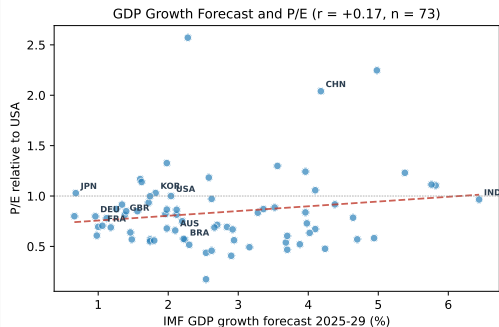


Country R&D/Assets vs P/E: $r = +0.41$,
 $t = 3.91$

Physical investment (CapEx): null ($r = +0.03$).
Not an accounting artifact: adding R&D back to earnings does not change the cross-country pattern. Not driven by IFRS vs GAAP differences.

Supply-side channel: high-P/E countries invest more in *intangibles*, not physical capital.

Not About Expected Growth



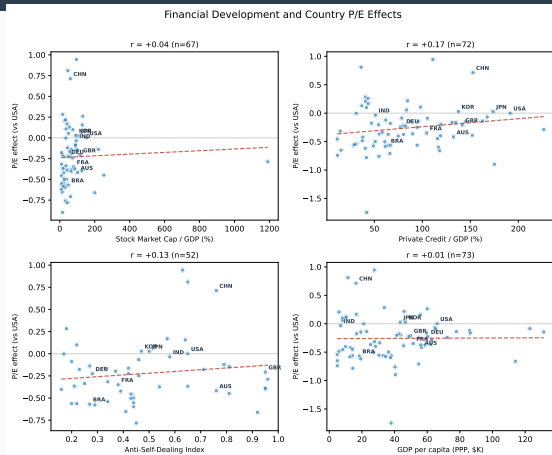
IMF 5-year GDP growth forecasts vs P/E:
 $r = +0.17$, $t = 1.42$ (null)

Also null: realized earnings growth ($r \approx 0$),
realized GDP growth ($r = +0.18$, $t = 1.56$).

Cross-country P/E differences are **not** driven
by expected growth rates. India (6.4%
forecast) has $P/E \approx USA$. Japan (0.7%)
matches Korea (1.8%).

Growth becomes marginally significant
($t = 1.69$) only as a third control alongside
R&D and savings.

Standard Institutional Variables Do Not Predict P/E



Market cap/GDP ($r = +0.04$), credit/GDP ($r = +0.17$), anti-self-dealing ($r = +0.13$), GDP per capita ($r = +0.01$): all near zero. La Porta et al. (2002) variables do not explain valuation levels.

Residual Country Effects After Controls

How much P/E variation do we explain?

Controls	σ_{resid}	Reduction
Raw country effects	0.394	—
+ Firm-level R&D	0.393	0.3%
R&D + Savings	0.315	20%
R&D + Sav + Growth	0.293	26%

Country-level R&D and savings explain 26% of cross-country P/E variation.

74% remains: large residual country effects persist.

Largest unexplained effects:

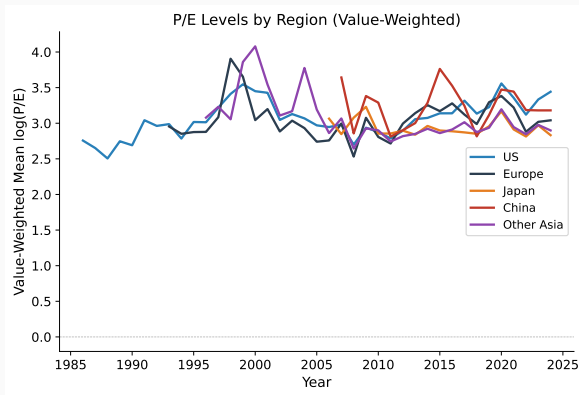
	Raw	Residual
CHN	+0.71	+0.50
IND	+0.42	+0.33
USA	0	−0.12
BRA	−0.56	−0.32
JPN	−0.01	+0.11

China's premium survives all controls.

Brazil's discount only partly explained.

Regional Patterns

Regional P/E Over Time

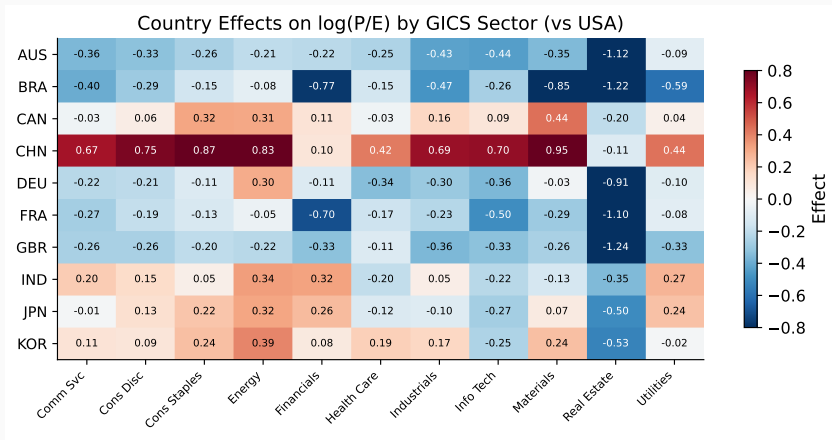


Key patterns:

- US P/E premium is recent and driven by large firms (tech)
- China P/E volatile, declining since 2015
- Japan near Europe

Value-weighted averages. VW gaps larger than EW — US premium concentrated in mega-caps.

Industry × Country P/E Effects



Country P/E effects by GICS sector. The China premium is broad-based across industries. Real estate shows large discounts in most countries.

Summary and Next Steps

Stylized Facts

1. **Country dominates P/E** — 65% of R^2 (Shapley), after controlling for size, leverage, earnings growth, industry. Country effects survive firm-level R&D control (σ drops by 0.3%).
2. **P/E dispersion is diverging** — $\times 2.7$ since early 2000s.
3. **Two robust predictors of country P/E effects:**
 - **R&D intensity** ($t = 5.89$) — intangible-intensive countries valued higher.
 - **National savings** ($t = 2.88$) — more capital seeking equity returns.

$R^2 = 0.45$ with growth control ($n = 65$). Both survive jointly. Growth alone is null.
4. **Large residual effects persist** — 74% unexplained. China premium, Brazil discount survive all controls. Not about growth, institutions, or financial development.

Research Questions

1. **Why is P/E dispersion diverging?** ($\times 2.7$) — US tech? Post-GFC monetary policy? Capital flows?
2. **Can the savings channel be formalized?** — Savings/GNI predicts P/E and is orthogonal to ROA. A model where savings rates affect discount rates?
3. **Do low-P/E countries earn higher returns?** — Bekaert–Harvey, Fama–French international. If so, valuation gaps reflect risk premia.
4. **What explains China's P/E premium?** — Highest P/E among large countries. Retail investors? Capital controls? Savings rate (46% of GNI)?

Related Literature

Country vs industry effects:

- Heston–Rouwenhorst (1994, *JFE*) — country dominates returns. We extend to *levels*.
- Griffin–Karolyi (1998, *JFE*) — industry effects vary across sectors.

Institutions and investor protection:

- La Porta et al. (2002, *JF*) — investor protection raises Tobin's Q. Our results: weak for P/E.
- Hail–Leuz (2006, *JAR*) — institutions reduce cost of equity by up to 200 bps.
- Stulz (2005, *JF*) — “twin agency problems” prevent equalization.

Savings and asset prices:

- Abel (2003, *Brookings*) — effect of baby boomers' saving on asset prices.
- Caballero et al. (2008, *JME*) — global imbalances and asset price inflation.

Appendix

Full List of Tested Predictors

Variable	r with P/E	t -stat	N
R&D / Assets	+0.48***	4.73	76
Savings / GNI	+0.34***	3.13	68
ROA	-0.39	-3.75	76
<i>after R&D control</i>	-0.08	-0.76	76
Private Credit / GDP	+0.17	1.30	72
Pension Fund Assets / GDP	+0.14	1.16	56
IMF GDP growth forecast	+0.13	1.07	73
Anti-Self-Dealing Index	+0.13	0.92	52
Stock Market Cap / GDP	+0.04	0.29	67
GDP per capita (PPP)	+0.01	0.09	73
Realized earnings growth	-0.00	-0.04	80
CapEx / Assets	+0.03	0.24	78

ROA appears significant alone but is absorbed by R&D. The ROA-P/E correlation is mediated by intangible investment intensity.

Market Cap Construction (Detail)

US firms: `mkvalt` (pre-computed market value in USD millions). Available from 1986.

Non-US firms (Compustat Global): no pre-computed market value.

- Compute: $\text{Market Cap} = \text{cshoi}$ (shares outstanding) $\times$ `prccm` (month-end price)
- **Pitfall:** firms have multiple securities across exchanges and currencies

Example — SAP SE (`gvkey=103487`) in Dec 2022:

`iid=02W` `EUR` 96.39 ← correct

`iid=05W` `HUF` 38,620 ← wrong currency!

`iid=06W` `MXN` 2,009 ← wrong currency!

- **Solution:** use `prirow` field (primary security designation)

All data in local currency (millions). Cross-country *levels* absorbed by country FE.

Accounting Standards

Compustat Global reports accounting standard via acctstd:

Standard	Share
IFRS	59.7%
Domestic GAAP	39.3%
US GAAP	0.2%
Other / missing	0.8%

US firms: all US GAAP.

Country variation:

- EU: mandatory IFRS since 2005
- Japan: 93% domestic GAAP
- China: 96% domestic GAAP
- Korea: 99% IFRS (since 2011)

P/E is less sensitive to accounting standards than book-equity-based ratios. Country FE absorb time-invariant differences.